Assignment 1.4

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Task 1

```
# Prime Number Check (Without Functions)

num = int(input("Enter a number: "))

if num <= 1:
    print("Not a prime number")
else:
    is_prime = True

    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print("Prime number")
    else:
        print("Not a prime number")
```

Output

Input:

```css
Enter a number: 7
```

Output:

```typescript
Prime number
```

Task 2

```
# Optimized Prime Number Check

num = int(input("Enter a number: "))

if num <= 1:
    print("Not a prime number")
elif num == 2:
    print("Prime number")
else:
    is_prime = True

    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
```

```
    if is_prime:
        print("Prime number")
    else:
        print("Not a prime number")
```

Output

Enter a number: 29
Prime number

Task 3# Prime Number Check Using Functions

```
def is_prime(num):
    """
    Returns True if the number is prime, otherwise False.
    Uses optimized checking up to square root of the number.
    """
    if num <= 1:
        return False
    if num == 2:
        return True

    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            return False
```

```
    return True


# Main program
number =
int(input("Enter a number: "))

if is_prime(number):
    print("Prime number")
else:
    print("Not a prime number")
```

Output

```
Enter a number: 13
Prime number
```

Task 4

```
# Prime Number Check Using Functions

def is_prime(num):
    if num <= 1:
        return False

    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
```

```
            return False

    return True


number = int(input("Enter a number: "))

if is_prime(number):
    print("Prime number")
else:
    print("Not a prime number")
```

Output

```
Enter a number: 17
Prime number
```

Task 5

```
# Optimized Prime Check

num = int(input("Enter a number: "))

if num <= 1:
    print("Not a prime number")
else:
```

```python
is_prime = True

for i in range(2, int(num ** 0.5) + 1):
    if num % i == 0:
        is_prime = False
        break

if is_prime:
    print("Prime number")
else:
    print("Not a prime number")
```

Output

```
Enter a number: 17
Prime number
```